

KINGSTON MANLEY, Jamaica

WMO: 783970

Lat: 17.936N

Lon: 76.788W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.7	23.0	14.8	10.5	25.3	15.8	11.2	25.5	13.4	29.4	12.3	29.3	2.9	350	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	33.2	26.1	33.0	26.2	32.3	25.9	28.1	30.6	27.7	30.5	27.2	30.6	9.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.6	23.5	29.4	27.0	22.7	29.3	26.3	21.7	29.4	90.2	30.3	87.9	30.9	86.3	30.6	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.6	11.4	10.4	DB	21.0	34.8	1.4	1.0	20.0	35.6	19.1	36.1	18.3	36.7	17.3	37.4
			WB	17.1	29.1	1.3	0.7	16.1	29.6	15.3	30.0	14.6	30.5	13.7	31.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	26.9	26.8	27.1	27.9	28.6	29.3	29.5	29.5	29.3	28.7	28.2	27.4
	DBStd	1.48	1.06	0.97	1.03	0.95	1.04	1.19	1.05	1.08	1.15	1.21	1.09	1.21
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6669	523	469	531	537	576	581	606	604	578	581	544	539
	CDD18.3	3627	265	236	272	287	317	331	348	346	328	322	294	281
	CDH23.3	41110	2417	2145	2628	3129	3748	4209	4472	4418	4084	3802	3249	2809
	CDH26.7	15164	609	529	694	986	1392	1864	2030	1984	1745	1451	1070	809
Wind	WSAvg	4.4	3.9	4.2	4.2	4.3	4.9	6.1	5.5	4.8	4.0	3.6	3.3	3.7
Precipitation	PrecAvg	726	17	13	19	28	80	69	36	89	123	147	80	37
	PrecMax	1704	104	55	73	131	311	548	165	549	547	636	348	195
	PrecMin	177	0	0	0	0	1	0	0	0	1	4	2	0
	PrecStd	341	20	11	18	30	86	97	40	97	136	145	77	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	31.8	31.8	32.2	32.9	34.0	34.1	33.9	33.8	33.3	32.9	32.2
		MCWB	23.7	24.2	24.3	25.1	25.9	25.5	25.9	26.4	26.7	26.3	25.9	25.0
	2%	DB	31.0	30.9	31.0	31.2	32.1	33.0	33.2	33.0	33.0	32.5	32.0	31.2
		MCWB	23.9	24.2	24.4	25.0	25.8	26.0	26.0	26.5	26.5	26.2	25.6	24.7
	5%	DB	30.1	30.1	30.2	30.9	31.4	32.2	32.6	32.2	32.2	32.0	31.2	30.8
		MCWB	23.8	23.9	24.1	24.9	25.6	25.9	25.9	26.3	26.3	26.0	25.4	24.6
	10%	DB	29.5	29.2	29.8	30.2	31.0	31.9	32.0	32.0	31.8	31.1	30.8	30.0
		MCWB	23.5	23.4	23.9	24.5	25.5	25.8	25.8	26.2	26.2	25.9	25.2	24.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.1	26.6	26.7	27.2	28.1	28.2	28.2	28.7	28.7	28.1	27.9	26.9
		MCDB	28.4	29.3	29.2	30.1	30.5	30.6	30.9	31.2	30.7	30.4	29.8	29.3
	2%	WB	25.3	25.5	25.8	26.3	27.1	27.5	27.5	27.9	27.9	27.5	27.0	26.1
		MCDB	28.4	29.0	28.8	29.7	30.4	30.8	31.0	30.9	30.4	30.2	29.5	29.2
	5%	WB	24.8	24.8	25.2	25.8	26.6	26.9	26.9	27.4	27.4	27.0	26.5	25.5
		MCDB	28.4	28.6	28.7	29.4	30.1	30.9	31.0	30.8	30.5	30.2	29.5	29.0
	10%	WB	24.2	24.2	24.6	25.3	26.2	26.5	26.5	26.8	26.9	26.5	25.9	25.0
		MCDB	28.1	28.1	28.4	29.1	29.8	30.6	30.8	30.7	30.5	29.9	29.4	28.8
Mean Daily Temperature Range	5% DB	MDBR	5.8	5.8	5.6	5.2	5.0	5.0	5.3	5.2	5.2	5.3	5.3	5.7
		MCDBR	6.5	6.4	6.1	5.7	5.4	5.4	5.7	5.5	5.6	5.8	5.8	6.3
		MCWBR	3.5	3.7	3.5	3.3	3.2	3.2	3.5	3.4	3.4	3.2	3.2	3.4
	5% WB	MCDBR	6.0	6.1	5.8	5.3	5.1	5.0	5.3	5.2	5.3	5.5	5.5	5.8
		MCWBR	3.7	3.9	3.6	3.5	3.3	3.3	3.6	3.6	3.6	3.3	3.4	3.6
Clear-Sky Solar Irradiance	taub	0.386	0.388	0.391	0.425	0.455	0.507	0.509	0.472	0.431	0.395	0.392	0.383	
	taud	2.519	2.497	2.485	2.399	2.341	2.171	2.157	2.299	2.429	2.552	2.532	2.550	
	Ebn at Noon	899	915	919	884	846	796	795	832	870	897	886	892	
	Edh at Noon	100	107	112	123	128	151	153	134	117	100	98	95	
All-Sky Solar Radiation	RadAvg	4.53	5.28	5.90	6.33	5.99	6.03	6.28	6.00	5.64	4.88	4.49	4.24	
	RadStd	0.24	0.19	0.23	0.26	0.39	0.35	0.39	0.30	0.43	0.44	0.31	0.22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page